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UNIT 5 VARIATION AND SELECTION

The genetic code is contained within a chemical in the nucleus of the cell, called DNA. DNA is organised into units called genes. The discovery of the structure of DNA in 1953 was the beginning of a new discipline called molecular biology. Molecular biology looks at how the genetic code in the DNA affects the function of the cell. Molecular biology is closely linked with genetics, which concerns the visible effects of the genes on living organisms . This Unit starts with a look at how DNA forms the genetic code, and how it is passed from cell to cell when a cell divides. Chapter 18 deals with genetics – how the DNA shows itself in organisms. Chapter 19 describes the theory that underpins all of biology – Darwin's theory of evolution by natural selection. The last chapter in this Unit describes the principles of selective breeding.



16 CHROMOSOMES, GENES AND DNA

This chapter looks at the structure and organisation of genetic material, namely chromosomes, genes and DNA.

LEARNING OBJECTIVES

- Understand that the nucleus of a cell contains chromosomes on which genes are located
- Understand that a gene is a section of a molecule of DNA that codes for a specific protein

BIOLOGY ONLY

- Describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)
- Understand that an RNA molecule is single stranded and contains uracil (U) instead of thymine (T)
- Describe the stages of protein synthesis (transcription and translation), including the role of mRNA, ribosomes, tRNA, codons and anticodons

- Understand that mutation is a rare, random change in genetic material that can be inherited

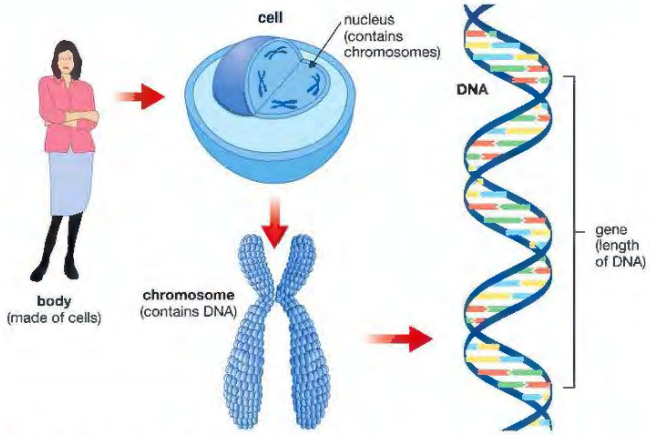
BIOLOGY ONLY

- Understand how a change in the DNA can affect the phenotype by altering the sequence of amino acids in a protein
- Understand how most genetic mutations have no effect on the phenotype, some have a small effect and rarely do they have a significant effect
- Understand that the incidence of mutations can be increased by exposure to ionising radiation (for example, gamma rays, x-rays and ultraviolet rays) and some chemical mutagens (for example, chemicals in tobacco)

- Know that in human cells the diploid number of chromosomes is 46 and the haploid number is 23
- Understand that the genome is the entire DNA of an organism
- Understand how genes exist in alternative forms called alleles, which give rise to different inherited characteristics.

DID YOU KNOW?

DNA is short for deoxyribonucleic acid. It gets the 'deoxyribo' part of its name from the sugar in the DNA molecule – deoxyribose, a sugar containing five carbon atoms.



▲ Figure 16.1 Our genetic make-up

The chemical that is the basis of inheritance in nearly all organisms is **DNA**. DNA is usually found in the nucleus of a cell, in the **chromosomes** (Figure 16.1). A small section of DNA that determines a particular feature is called a **gene**. Genes determine features by instructing cells to produce particular proteins which then lead to the development of the feature. So a gene can also be described as a section of DNA that codes for a particular protein.

DNA can replicate (make an exact copy of) itself. When a cell divides by mitosis (see Chapter 17), each new cell receives exactly the same type and amount of DNA. The cells formed are *genetically identical*.

BIOLOGY ONLY

THE STRUCTURE OF DNA

WHO DISCOVERED IT?

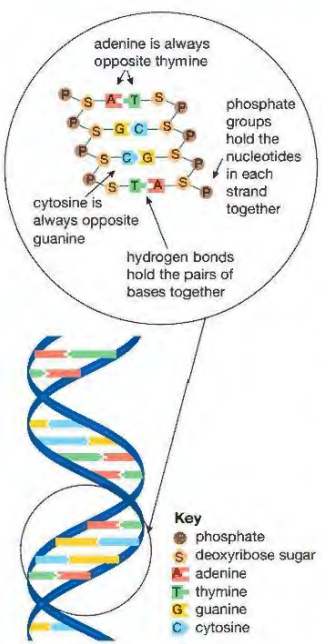
James Watson and Francis Crick, working at Cambridge University, discovered the structure of the DNA molecule in 1953 (Figure 16.2(a)). Both were awarded the Nobel prize in 1962 for their achievement. However, the story of the first discovery of the structure of DNA goes back much further than this. Watson and Crick were only able to propose the structure of DNA because of the work of others – Rosalind Franklin (Figure 16.2(b)) had been researching the structure of a number of substances using a technique called X-ray diffraction.



▲ Figure 16.2 (a) Watson and Crick with their double helix model



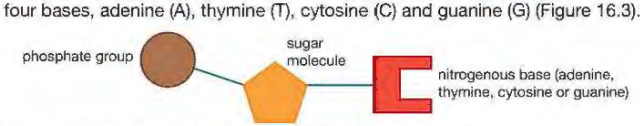
▲ Figure 16.2 (b) Rosalind Franklin (1920–1958)



▲ Figure 16.4 Part of a molecule of DNA.

Watson and Crick were able to use her results, together with other material, to propose the now-familiar double helix structure for DNA. Rosalind Franklin died of cancer in 1958 and so was unable to share in the award of the Nobel Prize.

A molecule of DNA is made from two strands of molecules called *nucleotides* (Figure 16.3). Each nucleotide contains a sugar molecule (deoxyribose), a phosphate group and a nitrogen-containing group called a **base**. There are four bases, adenine (A), thymine (T), cytosine (C) and guanine (G) (Figure 16.3).



▲ Figure 16.3 The structure of a single nucleotide.

Notice that, in the two strands (see Figure 16.4), nucleotides with adenine are always opposite nucleotides with thymine, and cytosine is always opposite guanine. Adenine and thymine are *complementary* bases, as are cytosine and guanine. Complementary bases always link or 'bind' with each other and never with any other base. This is known as the *base-pairing rule*.

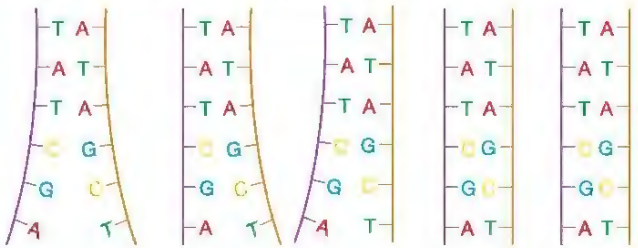
KEY POINT

One consequence of the base-pairing rule is that, in each molecule of DNA, the amounts of adenine and thymine are equal, as are the amounts of cytosine and guanine.

DNA is the only chemical that can replicate itself exactly. Because of this, it is able to pass genetic information from one generation to the next as a 'genetic code'.

DNA REPLICATION

When a cell is about to divide (see Mitosis, Chapter 17) it must first make an exact copy of each DNA molecule in the nucleus. This process is called **replication**. As a result, each cell formed receives exactly the same amount and type of DNA. Figure 16.5 summarises this process.



1 The polynucleotide strands of DNA separate.

2 Each strand acts as a template for the formation of a new strand of DNA.

3 DNA polymerase assembles nucleotides into two new strands according to the base-pairing rule.

4 Two identical DNA molecules are formed – each contains a strand from the parent DNA and a new complementary strand.

▲ Figure 16.5 How DNA replicates itself.

DID YOU KNOW?

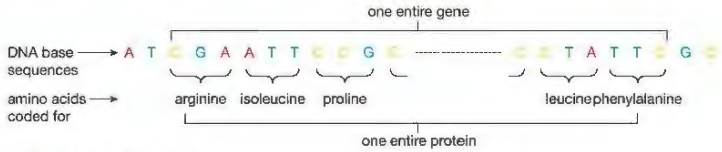
A 'template' is a pattern that can be used to make something. For example a dress template is a paper pattern for cutting out the material of a dress.

THE GENETIC CODE

Only one of the strands of a DNA molecule actually codes for the manufacture of proteins in a cell. This strand is called the **template strand.** The other strand is called the non-template strand.

Many of the proteins manufactured are enzymes, which go on to control processes within the cell. Some are structural proteins, e.g. keratin in the skin or myosin in muscles. Other proteins have particular functions, such as haemoglobin and some hormones.

Proteins are made of chains of amino acids. A sequence of *three* bases in the template strand of DNA codes for *one* amino acid. For example, the base sequence TGT codes for the amino acid cysteine. Because three bases are needed to code for one amino acid, the DNA code is a *triplet code*. The sequence of triplets that codes for *all* the amino acids in a protein is a gene (Figure 16.6).



▲ Figure 16.6 The triplet code.

KEY POINT

A gene is a section of a molecule of DNA that codes for a specific protein.

The triplets of bases that code for individual amino acids are the same in all organisms. The base sequence TGT codes for the amino acid cysteine in humans, bacteria, bananas, monkfish, or in any other organism you can think of – the DNA code is a *universal* code.

THE STAGES OF PROTEIN SYNTHESIS

DNA stays in the nucleus, but protein synthesis takes place in the cytoplasm. This means that for proteins to be made, the genetic code must be copied, and then transferred out of the nucleus to the cytoplasm. This is carried out by a different kind of nucleic acid called **ribonucleic acid (RNA)**.

DID YOU KNOW?

Ribose and deoxyribose are very similar in structure. Ribose contains an extra oxygen atom. Similarly, the bases uracil and thymine are very similar in structure.

There are three main differences between DNA and RNA:

- DNA is a double helix, RNA is a single strand
- DNA contains the sugar deoxyribose, RNA contains ribose
- RNA contains the base uracil (U) instead of thymine (T).

DID YOU KNOW?

There is a third type of RNA called ribosomal RNA (rRNA). Ribosomes are made of RNA and protein.

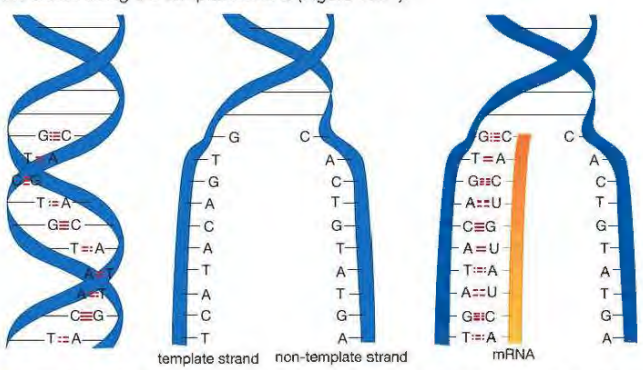
Two types of RNA take part in protein synthesis:

- **Messenger RNA (mRNA)**, which forms a copy of the DNA code
- **Transfer RNA (tRNA)**, which carries amino acids to the **ribosomes** to make the protein.

Protein synthesis takes place in two stages, called **transcription** and **translation**.

TRANSCRIPTION

Transcription happens in the nucleus. In a chromosome, part of the DNA double helix unwinds and 'unzips', so that the two strands separate, exposing the bases along the template strand (Figure 16.7).



▲ Figure 16.7 (a) The DNA double helix, showing some base pairs, (b) the two strands of the DNA have separated and unwound, (c) a short length of mRNA has formed.

KEY POINT

An RNA nucleotide consists of a ribose group, a phosphate and one of four bases (C, G, A or U).

The template strand of the DNA forms a framework upon which a molecule of mRNA is formed. The building blocks of the mRNA are RNA nucleotides. They line up alongside the template strand according to the complementary base-pairing rules (Table 16.1).

Table 16.1 Base-pairing rules in transcription

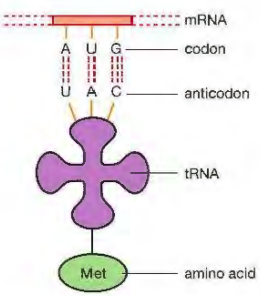
Base on DNA	Base on mRNA
G	C
C	G
T	A
A	U

One at a time, the RNA nucleotides link up to form an mRNA molecule. They form bonds between their ribose and phosphate groups, joining together to make the sugar–phosphate backbone of the molecule. When a section of DNA corresponding to a protein has been transcribed, the mRNA molecule leaves the DNA and passes out of the nucleus to the cytoplasm. It leaves through pores (holes) in the nuclear membrane. The DNA helix then 'zips up' again. Because of complementary base pairing the triplet code of the DNA is converted into a triplet code in the mRNA.

TRANSLATION

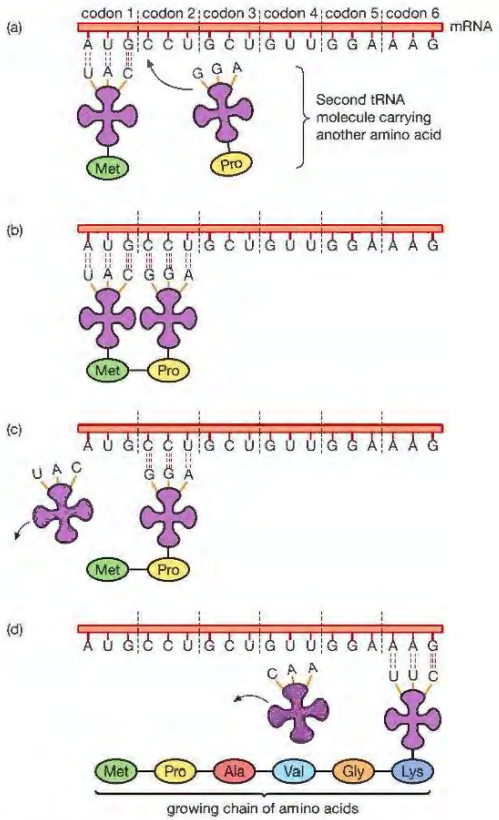
Converting the code in the mRNA into a protein is called translation. It takes place at the ribosomes. By this stage the code consists of sets of three bases in the mRNA (e.g. AUG, CCG, ACA). These triplets of bases are called **codons.** Each codon codes for a particular amino acid, e.g. CCU codes for the amino acid proline, and AUG codes for methionine.

The mRNA molecule attaches to a ribosome. Now the tRNA molecules begin their part in the process. Each has an **anticodon** of three bases at one end of the molecule, which is complementary to a particular codon on the mRNA. At the other end of the tRNA molecule is a site where a specific amino acid can attach (Figure 16.8). In other words there is a particular tRNA molecule for each type of amino acid. The tRNA molecule carries its amino acid to the ribosome, where its specific anticodon pairs up with the three bases of the corresponding mRNA codon.



▲ Figure 16.8 A tRNA molecule with the anticodon UAC, carrying the amino acid methionine.

This interaction between mRNA and tRNA is the basis of translation. The process is shown in Figure 16.9.



▲ Figure 16.9 Interaction between mRNA and tRNA is the basis of translation

It happens as follows:

- The first tRNA to bind at the mRNA does so at the 'start codon', which always has the base sequence AUG. This codes for the amino acid methionine.
- Another tRNA brings along a second amino acid. The anticodon of the second tRNA binds to the next codon on the mRNA.
- A bond forms between the methionine and the second amino acid.
- The first tRNA molecule is released and goes off to collect another amino acid.
- More tRNA molecules arrive at the mRNA and add their amino acids to the growing chain, forming a protein.
- At the end of the chain a 'stop codon' tells the translation machinery that the protein is complete, and it is released.

KEY POINT

Protein synthesis is a very energy-demanding process and uses a lot of the ATP made by the cell.

REMINDER

Summary of protein synthesis:

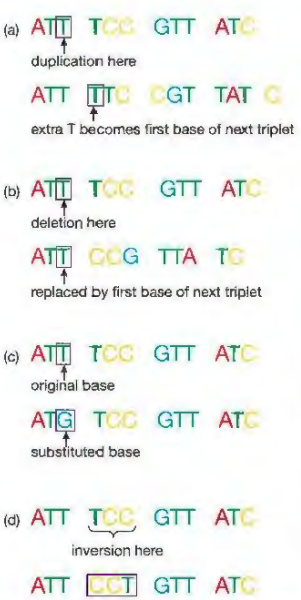
The order of bases in the template strand of the DNA forms the genetic code. The code is converted into the sequence of bases in the mRNA. In the cytoplasm the sequence of mRNA bases is used to determine the position of amino acids in a protein.

There are 20 different amino acids, so there need to be at least 20 different codons (and 20 different anticodons). In fact there are more than this, because some amino acids use more than one triplet code. For example the mRNA codons GGU, GGC, GGA and GGG all code for the amino acid glycine.

GENE MUTATIONS – WHEN DNA MAKES MISTAKES

KEY POINT

Double Award students only need to know that a mutation is a rare, random change in the genetic material that can be inherited. Students studying Biology will need to understand this section.



▲ Figure 16.10 Gene mutations (a) duplication, (b) deletion, (c) substitution, (d) inversion.

A **mutation** is a change in the DNA of a cell. It can happen in individual genes or in whole chromosomes. Sometimes, when DNA is replicating, mistakes are made and the wrong nucleotide is used. The result is a gene mutation and it can alter the sequence of the bases in a gene. In turn, this can lead to the gene coding for the wrong amino acid and therefore, the wrong protein. There are several ways in which gene mutations can occur (Figure 16.10).

In *duplication*, Figure 16.10 (a), the nucleotide is inserted twice instead of once. Notice that the entire base sequence is altered – each triplet after the point where the mutation occurs is changed. The whole gene is different and will now code for an entirely different protein.

In *deletion*, Figure 16.10 (b), a nucleotide is missed out. Again, the entire base sequence is altered. Each triplet after the mutation is changed and the whole gene is different. Again, it will code for an entirely different protein.

In *substitution*, Figure 16.10 (c), a different nucleotide is used. The triplet of bases in which the mutation occurs is changed and it *may* code for a different amino acid. If it does, the structure of the protein molecule will be different. This may be enough to produce a significant alteration in the *functioning* of a protein or a total lack of function. However, the new triplet may not code for a different amino acid as most amino acids have more than one code. (In this case, the protein will have its normal structure and function.)

In *inversions*, Figure 16.10 (d), the sequence of the bases in a triplet is reversed. The effects are similar to substitution. Only one triplet is affected and this may or may not result in a different amino acid and altered protein structure.

Mutations that occur in body cells, such as those in the heart, intestines or skin, will only affect that particular cell. If they are very harmful, the cell will die and the mutation will be lost. If they do not affect the functioning of the cell in a major way, the cell may not die. If the cell then divides, a group of cells containing the mutant gene is formed. When the organism dies, however, the mutation is lost with it; it is not passed to the offspring. Only mutations in the gametes or in the cells that divide to form gametes can be passed on to the next generation. This is how genetic diseases begin.

Sometimes a gene mutation can be an advantage to an individual. For example, as a result of random mutations, insects can become resistant to insecticides (see Chapter 15). Resistant insects obviously have an advantage over non-resistant types when an insecticide is being used. They will survive the insecticide treatment and reproduce. Their offspring will be resistant and so the proportion of resistant individuals will increase generation after generation. This is an example of **natural selection.** Bacteria can become resistant to antibiotics in a similar way (see Chapter 19).

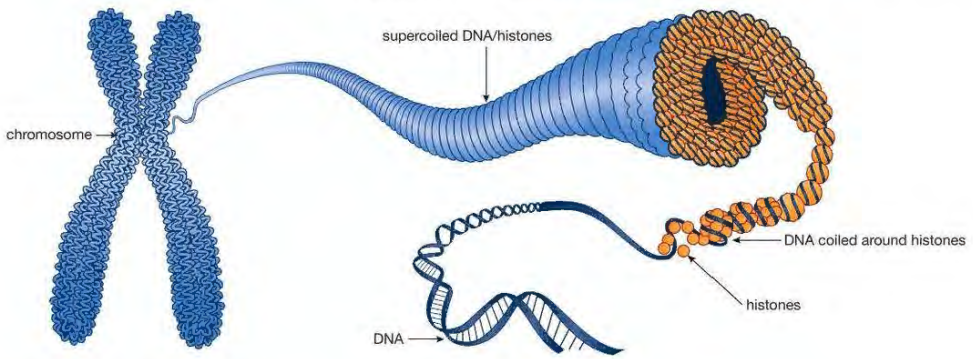
Gene mutations are random events that occur in all organisms. The rate at which they occur can be increased by a number of agents called **mutagens**. Mutagens include:

- ionising radiation (such as ultraviolet light, X-rays and gamma rays)
- chemicals including mustard gas and nitrous oxide, and many of the chemicals in cigarette smoke and the tar from cigarettes.

END OF BIOLOGY ONLY

THE STRUCTURE OF CHROMOSOMES

Each chromosome contains one double-stranded DNA molecule. The DNA is folded and coiled so that it can be packed into a small space. The DNA is coiled around proteins called **histones** (Figure 16.11).

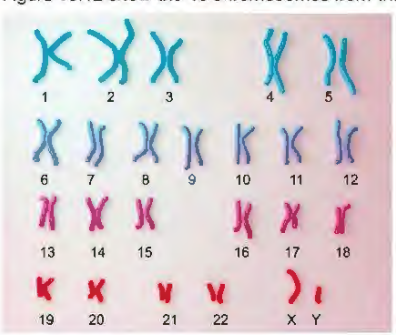


▲ Figure 16.11 The structure of a chromosome

Because a chromosome contains a particular DNA molecule, it will also contain the genes that make up that DNA molecule. Another chromosome will contain a different DNA molecule, and so will contain different genes.

HOW MANY CHROMOSOMES?

Nearly all human cells contain 46 chromosomes. The photographs in Figure 16.12 show the 46 chromosomes from the body cells of a human male.



◀ Figure 16.12 A man's chromosomes. One of each of the 22 homologous pairs are shown, along with the X and Y sex chromosomes. A woman's chromosomes are the same, except that she has two X chromosomes. A picture of all the chromosomes in a cell is called a karyotype.

REMINDER

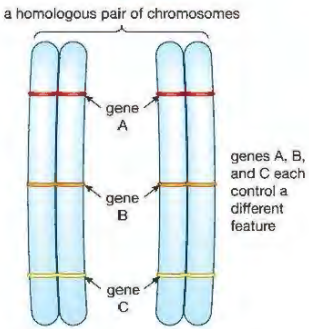
Red blood cells have no nucleus and therefore no chromosomes. (The lack of a nucleus means there is more room for carrying oxygen.)

KEY POINT

The X and the Y chromosomes are the **sex chromosomes**. They determine whether a person is male or female (see Chapter 18).

The chromosomes are not arranged like this in the cell. The original photograph has been cut up and chromosomes of the same size and shape 'paired up'. The cell from the male has 22 pairs of chromosomes and two that do not form a pair – the X and Y chromosomes. A body cell from a female has 23 matching pairs including a pair of X chromosomes.

Pairs of matching chromosomes are called **homologous pairs**. They carry genes for the same features, and these genes are arranged at the same positions and sequence along the chromosome (Figure 16.13). Cells with chromosomes in pairs like this are **diploid** cells.



▲ Figure 16.13 Both chromosomes in a homologous pair have the same sequence of genes.

Not all human cells have 46 chromosomes. Red blood cells have no nucleus and so have none. Sex cells have only 23 – just half the number of other cells. They are formed by a cell division called **meiosis** (see Chapter 17). Each cell formed has one chromosome from each homologous pair, and one of the sex chromosomes. Cells with only half the normal diploid number of chromosomes, and therefore only half the DNA content of other cells, are **haploid** cells.

When two gametes fuse in **fertilisation**, the two nuclei join to form a single diploid cell (a **zygote**). This cell has, once again, all its chromosomes in homologous pairs and two copies of every gene. It has the normal DNA content.

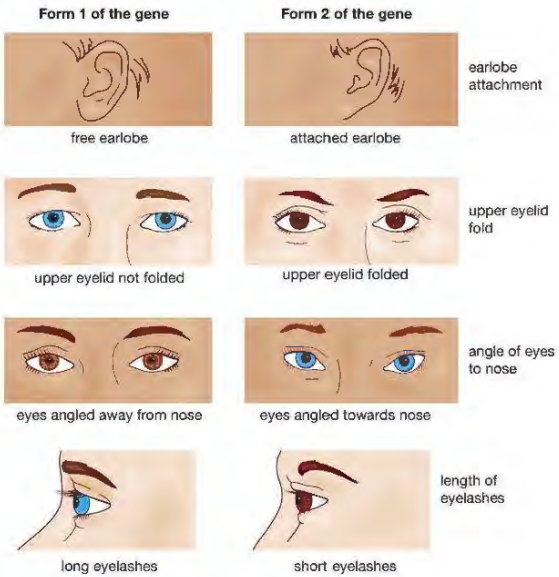
HOW MANY GENES?

The entire DNA of an organism (the amount present in a diploid cell) is known as its **genome**. The human genome is made up of about 3.2 billion base pairs. One of the surprise discoveries of modern molecular biology is that only a small fraction of the genome consists of protein-coding genes. For example, the human genome contains about 20 000-25 000 genes coding for proteins, which is only about 1.5% of the total DNA. The rest have other functions, or functions yet to be discovered! (See the 'Looking ahead' feature at the end of this chapter.)

GENES AND ALLELES

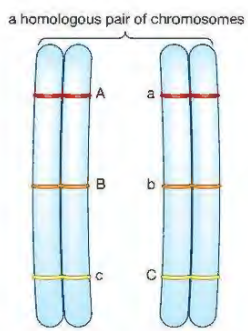
Genes are sections of DNA that control the production of proteins in a cell. Each protein contributes towards a particular body feature. Sometimes the feature is visible, such as eye colour or skin pigmentation. Sometimes the feature is not visible, such as the type of haemoglobin in red blood cells or the type of blood group antigen on the red blood cells.

Some genes have more than one form. For example, the genes controlling several facial features have alternative forms, which result in alternative forms of the feature (Figure 16.14).



▲ Figure 16.14 The alternate forms of four facial features

The gene for earlobe attachment has the forms 'attached earlobe' and 'free earlobe'. These different forms of the gene are called **alleles**. Homologous chromosomes carry genes for the same features in the same sequence, but the alleles of the genes may not be the same (Figure 16.15). The DNA in the two chromosomes is not quite identical.



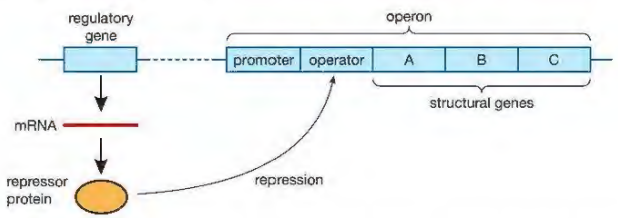
▲ Figure 16.15 A and a, B and b, and C and c are different alleles of the same gene. They control the same feature but code for different expressions of that feature.

Each cell with two copies of a chromosome also has two copies of the genes on those chromosomes. Suppose that, for the gene controlling earlobe attachment, a person has one allele for attached earlobes and one for free earlobes. What happens? Is one ear free and the other attached? Are they both partly attached? Neither. In this case, both earlobes are free. The 'free' allele is **dominant.** This means that it will show its effect, whether or not the allele for 'attached' is present. The allele for 'attached' is called **recessive.** The recessive allele will only show up in the appearance of the person if there is no dominant allele present. You will find out more about how genes work in Chapter 18.

LOOKING AHEAD – REGULATING GENES

You have seen how 'normal' gene coding for proteins (known as structural genes) make up only 1.5% of the genome. Some of the rest of the genome is DNA that regulates the action of the structural genes, switching them on and off. One way this happens is in regions of the DNA called operons.

An operon is a group of structural genes headed by a non-coding length of DNA called an operator, along with another sequence of DNA called a promoter. The promoter starts transcription by binding to an enzyme called RNA polymerase. Close to the promoter is a regulatory gene, which codes for a protein called a repressor. The repressor can bind with the operator, preventing the promoter from binding with RNA polymerase, and stopping transcription (Figure 16.16).



▲ Figure 16.16 An operon is a group of structural genes linked to an operator and a promoter. It is under the control of a regulatory gene.

The structural genes are in groups because they are related – e.g. they code for different enzymes in a metabolic pathway. Operons were first discovered in bacteria. At first we thought they only existed in prokaryotes, but molecular biologists have now found them in eukaryotic cells too.

CHAPTER QUESTIONS

More questions on DNA can be found at the end of Unit 5 on page 277.

BIOLOGY ONLY

SKILLS CRITICAL THINKING

1 Which of the following are components of DNA?

A deoxyribose, uracil and phosphate
B ribose, adenine and guanine
C deoxyribose, phosphate and adenine
D ribose, thymine and cytosine

2 Which of the following is the function of transfer RNA?

A transporting amino acids
B coding for the order of amino acids
C transcription of the DNA
D translation of the DNA

3 The base sequence for the same length of DNA before and after a gene mutation was as follows:

Before mutation: ATT TCC GTT ATC CGG
After mutation: ATT CCG TTA TCC GGA

Which type of mutation took place?

A duplication **B** deletion
C substitution **D** inversion

END OF BIOLOGY ONLY

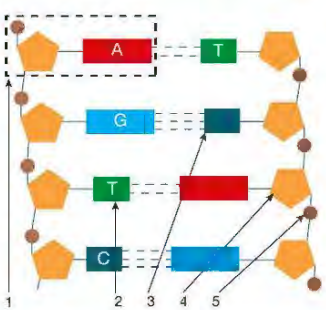
4 How many chromosomes are there in the body cells of a man?

A 23 pairs + XX **B** 23 pairs + XY
C 22 pairs + XX **D** 22 pairs + XY

BIOLOGY ONLY

SKILLS INTERPRETATION

5 The diagram represents part of a molecule of DNA.



a Name the parts labelled 1, 2, 3, 4 and 5.

SKILLS CRITICAL THINKING

b What parts did James Watson, Frances Crick and Rosalind Franklin play in discovering the structure of DNA?

SKILLS INTERPRETATION

c Use the diagram to explain the base-pairing rule.

END OF BIOLOGY ONLY

SKILLS CRITICAL THINKING

6 a What is:
i a gene
ii an allele?

b Describe the structure of a chromosome.

c How are the chromosomes in a woman's skin cells:
i similar to
ii different from those in a man's skin cells?

BIOLOGY ONLY

7 DNA is the only molecule capable of replicating itself. Sometimes mutations occur during replication.

a Draw a flow diagram to describe the process of DNA replication.

b Explain how a single gene mutation can lead to the formation of a protein in which:
i many of the amino acids are different from those coded for by the non-mutated gene
ii only one amino acid is different from those coded for by the non-mutated gene.

SKILLS ANALYSIS

8 Below is a base sequence from part of the template strand of a DNA molecule.

TAC CTC GGT CAT CCC

a How many amino acids are coded for by this base sequence?

SKILLS CRITICAL THINKING

b The sequence of the coding strand was transcribed to form mRNA. Write down the base sequence of this mRNA.

c Write down the corresponding base sequence of the *non-template* strand of the DNA.

d Copy and complete this description of the next stage in protein synthesis:

The mRNA base sequence is converted into the amino acid sequence of a protein during a process called _____ .
The mRNA sequence consists of a triplet code. Each triplet of bases is called a _____ . Reading of the mRNA base sequence begins at a _____ and ends at a _____ . Molecules of tRNA carrying an amino acid bind to the mRNA at an organelle called the _____ .

END OF BIOLOGY ONLY